

Problem 13

Based on the general term of the series, a_n , conclude that the following series diverges:

$$\sum_{n=1}^{\infty} \frac{n}{n+1}.$$

Solution

The Divergence Test (n-th Term Test)

Theorem: If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Contrapositive: If $\lim_{n \rightarrow \infty} a_n \neq 0$ (or if the limit does not exist), then the series $\sum_{n=1}^{\infty} a_n$ diverges. This is known as the **Divergence Test** or **n-th Term Test for Divergence**.

Step 1: Identify the General Term

The general term of our series is:

$$a_n = \frac{n}{n+1}.$$

Step 2: Calculate the Limit of the General Term

We need to evaluate:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{n+1}.$$

To find this limit, we divide both the numerator and denominator by n :

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{n/n}{(n+1)/n} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}}.$$

As $n \rightarrow \infty$, we have $\frac{1}{n} \rightarrow 0$, so:

$$\lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = \frac{1}{1 + 0} = 1.$$

Therefore:

$$\lim_{n \rightarrow \infty} a_n = 1 \neq 0.$$

Step 3: Apply the Divergence Test

Since $\lim_{n \rightarrow \infty} a_n = 1 \neq 0$, by the Divergence Test, the series:

$$\sum_{n=1}^{\infty} \frac{n}{n+1}$$

diverges.

Intuitive Explanation

For a series to converge, the terms must become arbitrarily small as n increases. In this case, the terms $a_n = \frac{n}{n+1}$ approach 1 as $n \rightarrow \infty$, meaning we are essentially adding approximately 1 infinitely many times:

$$\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \cdots \approx 1 + 1 + 1 + 1 + \cdots = \infty.$$

This clearly diverges to infinity.

Final Answer

Conclusion

The series $\sum_{n=1}^{\infty} \frac{n}{n+1}$ **diverges** because:

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0.$$

By the Divergence Test, if the limit of the general term is not zero, the series must diverge.